

Fabian Alexander Polnitzky

BSc BSc MSc

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and cosmology,
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Scientific Summary

I am working on **Bayesian inference** and **machine learning** methods and their applications in Astrophysics. Currently, I work on the **distribution of gas and dust** and their correlation in the **Milky Way** as part of the [mw-atlas](#) project, where I am using methods of **Information Field Theory** to reconstruct the spatial structure of our home galaxy. Furthermore, I work on the timescales of **protoplanetary disc evolution and dispersion**. From the observational side, I have experience with **all-sky observations** from optical, infrared to radio surveys.

Education

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- 10.2025 - 03.2029 **PhD**, RWTH Aachen University
Supervised by *Philipp Mertsch*
 - 10.2023 - 08.2025 **Master Astronomy**, University of Vienna
Master Thesis *Disk evolution in a single star-forming region*
supervised by *João Alves*, passed with honours
 - 10.2019 - 07.2023 **Bachelor Astronomy**, University of Vienna
Bachelor Thesis *Revisiting the ages of circumstellar disks*
supervised by *João Alves*, passed with honours
 - 10.2019 - 09.2023 **Bachelor Physics**, University of Vienna
Bachelor Thesis *Simulation and optimisation of microstrip antennas for propagating spin wave spectroscopy* supervised by *Andrii Chumak*, passed with honours

Publications

Polnitzky et al. 2026 *Precise determination of circumstellar disk lifetimes*, Astronomy and Astrophysics 707, A216; [10.1051/0004-6361/202554921](https://doi.org/10.1051/0004-6361/202554921)

Professional History

Research Internship on the automatic classification of young stellar objects' light curves at the European Southern Observatory under the supervision of *Amelia Bayo* and *Paula Sánchez Sáez*

Teaching

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- As **Teaching Assistant** at RWTH Aachen University
Theoretical Quantum Mechanics - Summer Semester 2025
 - As **Tutor** at the University of Vienna
 - Interstellar medium and star formation - Winter semester 2023 and 2024
 - Stars and planets - Winter semester 2023 and 2024
 - Astronomical lab course - Summer semester 2024
 - Observational lab course (Observatory practice) - Summer semester 2024
 - Astrophysics I (Part 1) - Summer semester 2022 and 2023
 - Astrophysics I (Part 2) - Winter semester 2022

Teaching

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Scholarships

Merit Scholarship of the University of Vienna awarded every year from 2020 to 2025.

Skills

Programming

Python - Proficient

Bash - Intermediate

HPC and **Sturm Workload Manager** - Intermediate

Git - Intermediate

SQL - Intermediate

TopCat - Intermediate

Aladin, **SAOImage DS9** - Beginner

Languages

German - Native

English - B2

Spanish - A1

Japanese - A1

Observations

Approved operator of the **Vienna Little Telescope** of the Department of Astrophysics, University of Vienna

Conferences

Plenary Talk at Star Formation Fields in Rethymno, Greece, in September 2026

Plenary Talk at Discs On the Exe in Exeter, UK, in July 2026

Parallel Talk at the DPG Spring Meeting in Erlangen, Germany, in March 2026

Part of the **LOC** and **Poster** at the Stellar Origins Conference in Vienna in September 2025

Extracurricular
Experience

Elected student representative for Astronomy from July 2021 to June 2025.